

Technology Stack

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Streamlit	Provides a simple, interactive, and user-friendly web interface for recording/uploading voice, viewing evaluation results, and displaying dashboards.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python	Provides fast processing and integrates AI libraries such as Whisper, Sentence Transformers, and Gemini AI for voice analysis and evaluation.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite	Lightweight relational database used to securely store transcripts, similarity scores, AI feedback, and evaluation history.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Deployment (Streamlit)	Simple deployment for project demonstration and testing with minimal configuration.

5	Version Control & CI/CD	GitHub	Used for version control, collaboration, source code management, and project backup.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	OpenAI Whisper Google Gemini API Sentence Transformers ReportLab SQLAlchemy	Whisper converts speech to text, Gemini AI generates feedback, Sentence Transformers calculate semantic similarity, ReportLab generates PDF reports, and SQLAlchemy manages database operations.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		